

SO(3) Equivariant Neural Networks

Lecture 2 Solutions

Features Have Geometric Types

Solutions to checkpoints, discussions, and notebook exercises

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1 Solutions to lecture questions

Question 2.1

R is the physical 3×3 rotation written in Cartesian vector coordinates. $D^{(\ell)}(R)$ is the matrix representing the same group element on a type- ℓ feature space. Its dimension is $(2\ell + 1) \times (2\ell + 1)$. For $\ell = 1$, $D^{(1)} = U_1 R U_1^\dagger$ is R in the spherical vector basis.

Question 2.2

The component counts are $2\ell + 1$: 1, 3, 5, 7 for $\ell = 0, 1, 2, 3$. Familiar interpretations are scalar or monopole for $\ell = 0$, vector or dipole for $\ell = 1$, quadrupole/symmetric-traceless tensor for $\ell = 2$, and an octupolar angular pattern for $\ell = 3$.

Question 2.3

The magnetic index $m = -\ell, \dots, +\ell$ labels the components of one irrep and is mixed by $D^{(\ell)}(R)$. A channel index labels multiple copies of the same irrep. The network designer chooses the number of channels; representation theory fixes the number of m components.

Question 2.4

Spherical components of a real physical tensor satisfy $(x_m^{(\ell)})^* = (-1)^m x_{-m}^{(\ell)}$. Therefore the complex entries are constrained and do not represent independent doubled degrees of freedom. The unitary basis change is invertible.

Question 2.5

Using $U_1^\dagger U_1 = I$,

$$D^{(1)}(R_a) D^{(1)}(R_b) = U_1 R_a U_1^\dagger U_1 R_b U_1^\dagger = U_1 (R_a R_b) U_1^\dagger = D^{(1)}(R_a R_b).$$

Thus a change of basis preserves the representation law.

Question 2.6

Checking only the imaginary residue asks whether the reconstruction is real, not whether it equals the original. A wrong real vector would pass. The correct test is

$$\max |\text{sph_to_cart}(\text{cart_to_sph}(v)) - v|.$$

2 Notebook reference checks

For the convention used in the series:

```
v_cart = rng.normal(size=(8, 3))
v_back = sph_to_cart(cart_to_sph(v_cart))
roundtrip = np.max(np.abs(v_back - v_cart))
```

```
D1 = U1 @ R @ U1.conj().T
```

```
route_a = cart_to_sph((R @ v_cart[0]))  
route_b = D1 @ cart_to_sph(v_cart[0])
```

Both residuals should be near 10^{-15} . For each implemented ℓ , also check

```
np.max(np.abs(D(1, Ra) @ D(1, Rb) - D(1, Ra @ Rb)))  
np.max(np.abs(D(1, R) @ D(1, R).conj().T - np.eye(2*1+1)))
```